

Empirical analysis on reliable cyber forensics analysis for cyber criminal investigation

by

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A Master's Thesis

submitted to the Florida Institute of Technology
at Florida Institute of Technology
in partial fulfillment of the requirements
for the degree of

Master of Science
in
Computer Science

Melbourne, Florida
Summer, 2025

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Abstract

Title:

Empirical analysis on reliable cyber forensics analysis for cyber criminal investigation

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The reliability of cyber forensic evidence acquisition is highly influenced by the underlying operating systems—Windows, Mac, and Linux—due to inherent variations in file system structures, encryption protocols, and the compatibility of forensic tools. Disk forensics, one of the most widely used techniques in digital investigations, faces distinct obstacles on each platform. Windows, with its predominantly NTFS and FAT file systems, typically supports reliable disk imaging and analysis through established tools such as FTK Imager and Autopsy/Sleuthkit. However, encryption features frequently pose challenges to evidence acquisition. Conversely, Linux environments, which rely on file systems like ext4 and XFS, generally offer greater transparency; yet, the transient nature of log retention often complicates forensic analysis efforts. In instances where anti-forensic strategies such as encryption and compression render traditional disk forensics insufficient, memory forensics emerges as an essential approach. While memory forensic methodologies demonstrate robustness across

Windows and Linux platforms through frameworks like Volatility, platform-specific difficulties persist. Memory analysis on Linux systems benefit from tools like LiME, snapshot utilities, and dd for memory acquisition. Nevertheless, data volatility during live memory acquisition on Linux can still present challenges. The research here seeks to systematically assess both disk and memory forensic acquisition techniques across varied samples representing Windows and Linux systems. By identifying the most effective combinations of forensic tools and configurations tailored to each operating system, the study aims to improve the accuracy and reliability of evidence collection. Furthermore, it critically evaluates the capabilities of current forensic tools, exposing a persistent research gap: the consistent assurance of forensic input reliability and footprint integrity. Through the integration of static analysis and temporal performance evaluations, this study emphasizes the importance of developing standardized forensic procedures and practices to ensure the integrity and admissibility of evidence across diverse platforms. The establishment of such standardised methodologies is imperative to maintaining the credibility and reliability of cyber forensic investigations in heterogeneous operating system environments.

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List of Symbols, Nomenclature or Abbreviations

ACPO Association of Chief Police Officers

ADS Alternate Data Streams

APFS Apple File System

AWS Amazon Web Services

C2 Command and Control

CAINE Computer Aided INvestigative Environment

dd Data Duplicator

DFRWS Digital Forensic Research Workshop

DLL Dynamic Link Library

E01 EnCase Evidence File Format

EXT4 Fourth Extended Filesystem

FAT File Allocation Table

FBI Federal Bureau of Investigation

FTK Forensic Toolkit

GB Gigabyte

HFS+ Hierarchical File System Plus

HVCI Hypervisor-Protected Code Integrity

IC3 Internet Crime Complaint Center

IoT Internet of Things

KB Kilobyte

LiME Linux Memory Extractor

LUKS Linux Unified Key Setup

MB Megabyte

MD5 Message Digest Algorithm 5

ML Machine Learning

MP3 MPEG Audio Layer III

MP4 MPEG-4 Part 14

NIST National Institute of Standards and Technology

NLP Natural Language Processing

NTP Network Time Protocol

NTFS New Technology File System

OS Operating System

PID Process Identifier

RAM Random Access Memory

RAT Remote Access Trojan

RFC Request for Comments

SANS SIFT SysAdmin, Audit, Network, Security - SANS Investigative Forensic Toolkit

SEC Securities and Exchange Commission

SHA Secure Hash Algorithm

SIP System Integrity Protection

T0 Time Zero (Acquisition without malware)

T1 Time One (Immediate acquisition after malware)

T2 Time Two (24 hours after malware)

T3 Time Three (72 hours after malware)

TCC Transparency, Consent, and Control

TCP Transmission Control Protocol

UDP User Datagram Protocol

UK United Kingdom

US United States

US-LATT United States Live Acquisition and Triage Tool

USB Universal Serial Bus

VBS Virtualization-Based Security

VM Virtual Machine

VSS Volume Shadow Copy Service

XFS X File System

Acknowledgements

I would like to express my sincere gratitude to my advisor, Dr. Sneha Sudhakaran, for her invaluable guidance, expertise, and unwavering support throughout this research journey. Her profound knowledge in the field of cyber forensics and computer security has been instrumental in shaping this thesis. I am deeply thankful for her patience, motivation, and for challenging me to push the boundaries of my understanding and capabilities.

I extend my appreciation to my committee members, Dr. Christian Sonnenberg and Dr. Neji Mensi, for their insightful comments, constructive feedback, and hard questions which encouraged me to widen my research perspectives. Special thanks to Dr. Brian Lail for his support as Department Head.

I am grateful to the Florida Institute of Technology for providing an excellent academic environment and resources necessary for conducting this research. My sincere thanks also go to the faculty and staff of the Department of Electrical Engineering and Computer Science for their assistance and encouragement.

Finally, I must express my profound gratitude to my family for providing me with unfailing support and continuous encouragement throughout my years of study, through the process of researching and writing this thesis, and throughout my life in general. This accomplishment would not have been possible without them.

Thank you all.

Dedication

This thesis is dedicated to my parents, whose unwavering love, support, and sacrifices have made this achievement possible. Your belief in me has been my strength throughout this journey.

To my supervisor who inspired my interest in cybersecurity and digital forensics, your wisdom and guidance have shaped my academic path and professional aspirations.

To my friends and colleagues who provided encouragement, technical discussions, and much-needed breaks during the research process, made this challenging work enjoyable.

And finally, to all the cybersecurity professionals and digital forensic investigators working tirelessly to combat increasingly sophisticated cyber threats, may this research contribute, even if modestly, to your important work in making our digital world more secure.

Chapter 1

Introduction

1.1 Understanding Cyber Forensics and its Critical Role

Digital forensics, often referred to as cyber forensics, is a specialized field within forensic science dedicated to recovering, investigating, and analyzing digital evidence from electronic devices and networks. This discipline has advanced rapidly in response to the growing adoption of digital technologies and the corresponding surge in cybercrime. With the expansion of global Internet usage, cybercrime has seen an alarming increase. The FBI's Internet Crime Complaint Center (IC3) reported 880,418 complaints in 2023, resulting in financial losses exceeding \$12.5 billion up from \$3.5 billion in 2019 [1, 2]. As of February 20, 2025, over 5 billion individuals worldwide, roughly 67.9% of the global population, were Internet users [3, 4]. This widespread connectivity has broadened opportunities for cyber criminals, underscoring the need for sophisticated forensic methods to effectively combat these evolving threats.

1.1.1 Categories of Digital Crime

In response to the growing complexity and volume of cybercrime, investigators have developed specialized branches of cyber forensics. Among these, computer forensics stands as a cornerstone, focusing primarily on desktop operating systems [5]. Desktop computers often serve as key repositories of evidence in cases of corporate espionage, financial fraud, child exploitation, and insider threats [6]. Beyond computer forensics, other vital branches include network forensics (analyzing traffic and infrastructure), mobile device forensics, cloud forensics, and memory forensics [6, 7]. Desktop systems, in particular, remain indispensable in investigations of corporate data breaches and financial crimes due to their role in storing critical information [7]. Although mobile devices have proliferated, desktops continue to be central in forensic efforts involving corporate fraud and insider threats [6, 7].

1.2 Significance and Reliability of Cyber Forensic Analysis

The importance of cyber forensic analysis is underscored by the substantial backlog of digital evidence that is awaiting examination by law enforcement agencies. In the UK, **over 21,000 digital devices** including mobile phones, tablets, and computers remain unexamined across police forces, with some cases delayed for **over four years** while evidence is processed [8, 9]. This backlog has severe implications for criminal justice, particularly in sensitive cases like sexual violence, where delays exacerbate trauma for victims and defendants awaiting resolution. Digital evidence is now involved in over 90% of criminal cases, ranging from homicides to financial fraud, reflecting its indispensable role in modern investigations [10, 11].

However, resource constraints plague agencies globally:

- 51% of digital forensic examiners report rising case backlogs [12].
- 66% of devices submitted to labs are locked, requiring advanced tools and expertise to access [12].
- Prosecutors face 100 to 500 cases in average backlogs, with 10% exceeding 1,000 cases [13].
- Investigators spend 45 hours per case reviewing digital evidence, often using outdated methods like USB drives or paper-based systems [13].

1.3 Trustworthiness, Credibility and Authenticity Requirements

The credibility of cyber forensic analysis depends on several interconnected factors. In 2007, Garfinkel highlighted the need for a methodological rigor, technical accuracy, reproducibility of results, and adherence to established legal standards [14]. The National Institute of Standards and Technology (NIST) emphasizes that forensic credibility requires maintaining an unbroken chain of custody, employing validated tools, and implementing comprehensive documentation procedures throughout the investigative process [15, 16]. Research by NIST’s Scientific Foundation Review highlights that digital forensic methods are based on established computer science principles but face challenges due to rapid technological advancements and gaps in peer review processes [17]. Additionally, Alec Noland reported in a recent study that the lack of universal standards in digital forensics contributes to inconsistent results, with defense attorneys frequently questioning the reliability of evidence acquisition methods in court [18].

These lack of standards are mentioned in works that includes all kinds of forensic analysis across different operating systems[19]. Furthermore, ensuring trustworthiness and authenticity in cyber forensic analysis requires validation of digital evidence to confirm it has not been altered or tampered with, as well as the adoption of universal standards to promote consistent and reliable results. Comprehensive peer review processes, adherence to best practices, and continuous advancements in forensic methodologies are essential to maintaining integrity, trust, and authenticity in forensic investigations.

1.4 Forensically sound practices for avoiding Flawed Cyber Forensic Practices

A significant challenge confronting cyber forensic practitioners is the rapidly changing technological landscape. According to the National Institute of Standards and Technology (NIST), digital forensic methods must continuously adapt to keep pace with evolving encryption protocols, emerging storage technologies, and sophisticated anti-forensic measures [17]. This constant state of flux often results in compatibility issues for forensic tools after major operating system updates. For instance, in 2022, approximately 43% of desktop operating system forensic tools experienced significant compatibility problems following updates, as reported by the Infosec Institute [20]. Anti-forensic strategies such as advanced encryption and data obfuscation add further complexity, compelling forensic practitioners to innovate and refine their methodologies to ensure reliable evidence acquisition [21]. These challenges highlight the necessity for ongoing validation and adaptation of forensic practices to maintain their credibility in court proceedings [22, 17]. The importance of credible cyber forensic analysis cannot be overstated, as digital evidence is pivotal in resolving modern crimes. However, errors in forensic procedures can have severe consequences, including wrongful

convictions and miscarriages of justice. Mishandled digital evidence has the potential to allow the actual perpetrators to escape accountability. For example, the Amanda Knox case revealed how improper forensic handling and evidence cross contamination contributed to a wrongful conviction that was eventually overturned [23]. Similarly, the David Camm case illustrated the dangers of flawed forensic practices, leading to years of incarceration for an innocent individual while the true perpetrators remained free [24]. These instances underscore the critical need for rigorous forensic standards to safeguard both the accuracy of legal judgments and the fairness of the judicial process.

Beyond individual cases, unreliable forensic practices can have widespread consequences, eroding public trust in both law enforcement agencies and judicial institutions. When digital evidence is deemed inadmissible or untrustworthy due to procedural failings, it undermines prosecution efforts, delays the delivery of justice, and increases both the financial and reputational costs associated with legal proceedings [25]. Victims of cybercrime may lose confidence in the legal system if they perceive justice as being delayed or denied because of forensic shortcomings. Given that digital evidence now plays a role in approximately 90% of criminal cases [6], ensuring reliable forensic methods is not merely an operational imperative it is essential to maintaining the integrity and public trust of the justice system as a whole. A flawed forensic analysis or misjudgment can result in falsified evidence, which not only endangers an innocent person's life and reputation but also allows the true perpetrator to escape justice.

1.5 Reliable Methods for Cyber Forensic Investigations

To achieve credible results in cyber forensic investigations, practitioners must employ structured methodologies that adhere to established best practices and leverage

advanced tools. For consistent and application specific analysis, the selection of appropriate tools is crucial [26, 27, 19]. Reliable methodologies start with understanding the context of the crime scene. Investigators must identify the type of crime such as data breaches, insider threats, or financial fraud and determine what information may be missing from the larger narrative [28]. A thorough mapping of all digital devices at the crime scene is critical, including computers, IoT devices like smart home systems, mobile phones, security cameras, network infrastructure, USB drives, and cloud storage platforms. Such comprehensive preparation ensures no vital evidence is overlooked during analysis [29].

Evidence acquisition must be handled meticulously to preserve its integrity. Forensic imaging tools such as FTK Imager or EnCase are used to create exact duplicates of storage media, ensuring original data remains unaltered [30]. Volatile data such as running processes or active network traffic should be captured immediately with tools like Wireshark, as it may be lost once systems are powered down [31]. Cybercriminals' anti-forensic measures, including encryption and steganography, require investigators to deploy advanced techniques to uncover hidden information. Reverse steganography methods can reveal concealed data in files or images by analyzing hash values and file attributes [32]. Additionally, data carving techniques allow recovery of partially deleted files by examining fragmented data stored elsewhere on the device.

Following evidence collection, rigorous analysis and validation steps are essential to ensure reliability. Timeline reconstruction correlates timestamps from disk images, registry entries, and network logs to piece together a chronological sequence of events [6]. Hash analysis helps maintain file integrity by verifying that data has not been tampered with. Live memory analysis provides insight into active processes and encryption keys at the time of the crime by capturing data from RAM [31].

Reliable investigations involve gathering evidence from diverse sources to build a

complete picture. Disk imaging preserves exact duplicates of hard drives, ensuring the integrity of original data [30]. System logs reveal critical information about user activity and access patterns. Investigating IoT devices, such as smart cameras, and cloud platforms requires specialized tools like Cellebrite or Oxygen Forensics to maintain cryptographic integrity [33]. Detecting anti-forensic techniques ensures that attempts at obfuscation or deletion do not obscure critical evidence. Building on the preceding analysis, we aim to identify reliable and effective forensic methods by conducting an empirical evaluation across a range of simulated case studies.

1. **Research Objective 1:** To systematically evaluate the effectiveness of different forensic techniques across simulated case studies, identifying those that yield the most reliable and trustworthy results.
2. **Research Objective 2:** To develop a set of best practices for assessing the credibility and accuracy of forensic methods through empirical analysis, ensuring consistency in real world applications.

In this thesis, chapter 2 explains literature review, Chapter 3 highlights the simulation of case studies. Chapter 4 highlights the results of our case studies and Chapter 5 concludes with Conclusion and Future work.

Chapter 2

Literature Survey

2.1 Cross-Platform Forensic Analysis for Cyber Criminal Investigations

Digital, computer, and cyber forensics have emerged as critical disciplines in modern criminal investigations, driven by the widespread proliferation of computing devices. This growth has introduced both opportunities and challenges for investigators. As cybercrime becomes increasingly sophisticated, forensic methodologies must evolve to address the diverse array of operating systems encountered in investigations. This literature survey focuses on empirical approaches to forensic analysis in the two PC operating systems, Windows, Linux, and also Mac OS for future work reference, with a specific emphasis on the challenges posed by malware infections.

The importance of cross-platform forensic proficiency has increased significantly as cyber criminals increasingly employ multiple operating systems in their operations. Recent findings suggest that approximately 43% of sophisticated cyber attacks involve multiple operating systems, either as targets or as attack platforms [34]. This multi

platform strategy presents distinct challenges for forensic investigators, who must possess expertise across various operating environments, each with unique file systems, memory management practices, and evidence artifact locations.

This survey synthesizes current research, methodological frameworks, and empirical findings to build a comprehensive understanding of reliable forensic analysis techniques that are applicable in diverse operating systems. Particular emphasis is placed on memory dumps and disk images as primary evidence sources, with close examination of how the presence of malware impacts the acquisition and analysis of these forensic artifacts. Commercial forensic tools are prevalent in the field of digital forensics, providing comprehensive solutions for evidence acquisition and analysis. These tools vary in their cross-platform capabilities and overall effectiveness. For instance, EnCase and X-Ways Forensics are highly regarded for their robust performance on Windows systems, excelling in recovering registry artifacts and analyzing NTFS volumes. On MacOS, tools like BlackLight are tailored to handle APFS disks, but sometimes encounter difficulties when working with FileVault-encrypted volumes. Magnet AXIOM, a widely used cross-platform solution, is notable for its advanced timeline analysis features, which enable artifact correlation across multiple operating systems [35, 36].

One common drawback of commercial forensic tools is their delay in supporting new operating system versions. Studies have shown that it often takes several months for these tools to fully adapt to major updates, with MacOS updates typically experiencing the longest delays compared to Windows and Linux distributions. This delay can impede timely investigations, particularly when dealing with cutting-edge technologies or recently released operating systems [37].

2.1.1 Open-Source Forensic Tools

Open-source forensic tools are becoming increasingly popular as cost-effective and flexible alternatives to commercial solutions. Tools such as The Sleuth Kit offer strong support for Linux file systems like EXT4, though they often lack comprehensive compatibility with MacOS's APFS. In cloud environments, Google Cloud Forensics Utils integrates open-source frameworks such as Plaso and Log2Timeline to streamline log analysis in hybrid environments. For memory forensics, Volatility has established itself as a leading framework, providing robust capabilities for examining memory artifacts on both Windows and Linux platforms [38, 39].

Open-source solutions frequently demonstrate faster adaptation to emerging technologies than their commercial counterparts, driven by active community contributions. This adaptability makes them particularly valuable for investigating unusual configurations, addressing novel file systems, and responding to newly developed platforms [32, 37]. The above literature has helped us to get an idea of how a forensic investigator should be efficient in identifying the best tool for acquisition and analysis in cases involving different platforms.

Digital forensic investigations are based on established frameworks to ensure systematic and methodical handling of evidence. Among the most prominent is the DFRWS Investigative Model, which comprises six well-defined phases: identification, preservation, collection, examination, analysis, and presentation [40]. This framework emphasizes scientific rigor and legal admissibility, serving as a cornerstone for modern investigative methodologies [41]. Furthermore, the ACPO Good Practice Guide builds on these phases, placing particular emphasis on chain-of-custody protocols and the validation of forensic tools [42].

The National Institute of Standards and Technology (NIST) offers another widely accepted framework that simplifies the investigative process into four core phases: col-

lection, examination, analysis, and reporting [41].

2.1.2 Scene Preservation and Initial Response

Effective initial response protocols are essential for preserving both physical and digital crime scenes. Key procedural steps include:

- **Documenting Connections:** Taking photographs of device ports and network configurations helps retain information about the state of volatile data (such as RAM and active processes) [42].
- **Isolating Suspects:** Preventing suspects from accessing devices ensures that anti-forensic actions such as remote wiping or data destruction are thwarted [42].
- **Using Write Blockers:** Employing hardware write blockers, such as Tableau TDs, safeguards evidence integrity by preventing any alterations during data acquisition [43].

The US-LATT (United States Live Acquisition and Triage Tool) methodology categorizes data based on their volatility[44], allowing investigators to prioritize their collection efforts [45]:

- **Highly Volatile Data:** RAM and active processes should be captured immediately, using tools such as Belkasoft, FTK Imager, or RAM Capturer for Windows environments [45].
- **Moderately Volatile Data:** Items like web history and registry events can be further analyzed after initial memory acquisition, often utilizing Volatility[44] plugins [45].

2.2 Acquisition of Evidential Data in Trusted Environment

Once we identify that the forensically acquired dump has a role in antifoensic data, a forensic investigator must acquire runtime or volatile data. This forces effective acquisition and analysis of volatile data for effective evidence recovery. The acquisition of volatile data is one of the most critical and technically challenging steps in digital forensics. Preserving ephemeral evidence before it is lost is paramount, and memory forensics plays a central role in this effort. By analyzing system RAM, investigators can recover valuable information that is not present in persistent media. Tools such as Belkasoft Live RAM Capturer are widely used in Windows systems due to their ability to operate at the kernel level and bypass antidebugging or antidumping measures, ensuring reliable memory extraction [46, 47].

For Linux, LiME (Linux Memory Extractor) provides full memory captures while limiting interactions between user and kernel space processes, thus maintaining forensic soundness. However, LiME can face challenges when compiling kernel modules in various Linux distributions [48, 49]. On MacOS, tools such as OSXPmem and MacQuisition are commonly used for volatile memory capture. Although effective, these tools can encounter difficulties with Apple’s advanced memory protection features, sometimes resulting in system crashes that risk data loss if not properly managed [50, 51].

Delays in acquiring data increase the chance of losing critical artifacts, as even minimal tool activity can modify memory contents. Despite these complications, volatile memory analysis remains indispensable, often uncovering encryption keys, login credentials, and other vital evidence [46, 47, 51].

Documenting every step of evidence handling is essential to ensure its legal admissibility. The chain of custody involves several key elements [43]:

Hash verification is a cornerstone of the verification of evidence integrity. Although MD5 and SHA-1 are still widely used, studies highlight their susceptibility to collision attacks, triggering a shift towards stronger hashing algorithms such as SHA-256 [43]. Adopting these more robust algorithms is recommended for verifying all digital evidence, regardless of the source of the operating system. Specialized forensic distributions, including SANS SIFT, Kali Linux, and CAINE, have gained prominence for creating standardized analysis environments. Each distribution offers unique strengths: SIFT is known for its broad compatibility with Windows, Linux, and MacOS evidence, while CAINE excels at integrating legal documentation workflows, but provides limited support for MacOS artifacts. Cloud-based platforms are emerging as essential tools in digital investigations. Solutions like Cado Security automate evidence collection in cloud environments such as AWS and Azure, significantly reducing analysis time compared to traditional methods. [32, 6].

Timeline reconstruction tools like **Magnet AXIOM** correlate events between platforms to establish comprehensive attack chronologies. Challenges such as clock-synchronization discrepancies between systems (e.g., Windows vs. Linux) complicate cross-platform timelines but can be mitigated through rigorous NTP implementation. Visualization techniques such as temporal heat maps improve the recognition of attack patterns, reducing analysis time, and enhancing investigative efficiency [52]. Correlating digital artifacts across platforms is essential for modern investigations. Tools such as **Cado Security** map cloud activity to on-premises evidence, linking AWS/Azure logs with local user actions. Graph-based analysis models files, processes, and users as nodes with interactions as edges, revealing hidden connections in multi-platform attacks. Behavior-based correlation focuses on attacker objectives rather than platform-specific methods, improving the detection of advanced threats tailored to specific environments [53].

Chapter 3

Design Methodology

3.1 Environment Set up

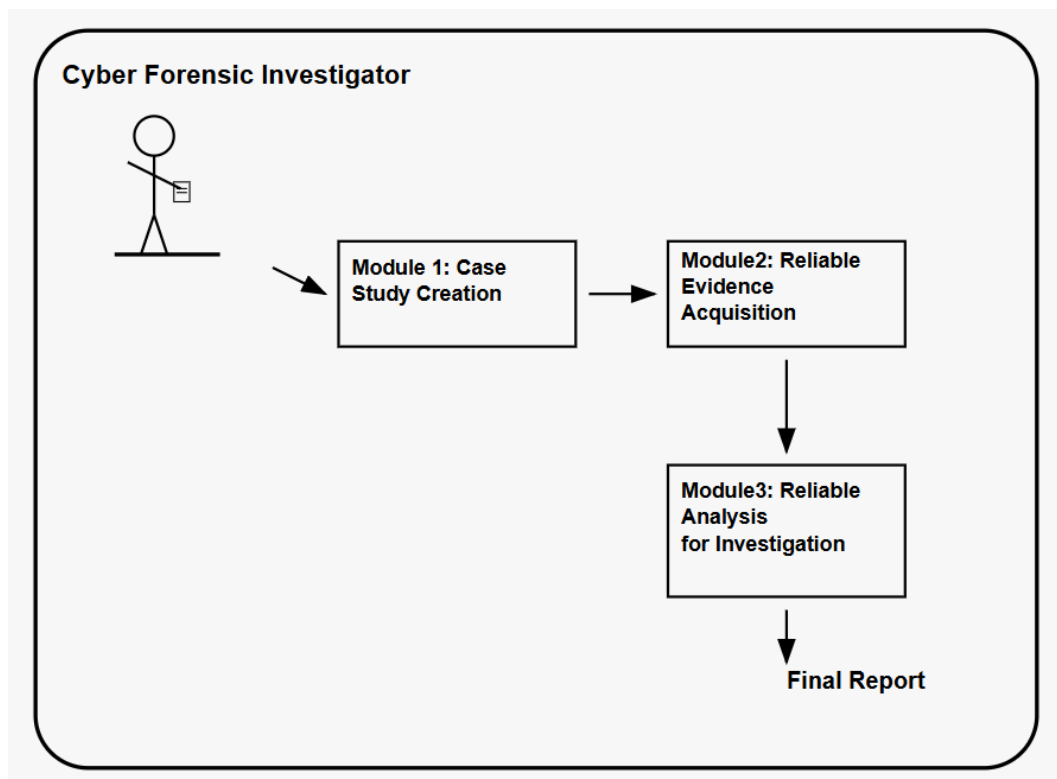


Figure 3.1: Experimental Setup

The workflow of this research is explained in Figure 3.1. The overall design focuses mainly on an empirical experimental best practice designed to evaluate forensic analysis capabilities in two widely used operating systems. We use virtual machine environments to simulate Windows and Linux systems, enabling controlled testing of forensic techniques. The primary goal is to establish a standardized methodology for assessing the reliability and effectiveness of forensic tools and identifying robust techniques suitable for diverse platforms. The experiment specifically focuses on two types of forensics: disk forensics and memory forensics. We investigate how variables such as system configuration, malware infection scenarios, and time taken - influence the quality and integrity of recovered evidence. By systematically measuring evidence degradation and assessing the impact of system resource availability, the research addresses practical challenges encountered in real-world cyber investigations. The empirical study consists of 3 modules:

1. Module 1: Case Study Creation
2. Module 2: Reliable Evidence Acquisition
3. Module 3: Reliable Analysis for Investigation

3.1.1 Module 1: Case Study Creation

The study selected four operating systems with two system memory settings as shown in table 3.1 that represent the most widely used desktop environments today.

Operating System	Version	RAM Configuration	Storage Capacity
Windows	Windows 11	4GB / 8GB	80GB
Windows	Windows 10	4GB / 8GB	80GB
Linux	Ubuntu 24	4GB / 8GB	80GB
Linux	Ubuntu 20	4GB / 8GB	80GB

Table 3.1: Operating Systems and Their Configurations

The study used operating systems that collectively account for approximately 90% of current desktop usage worldwide, based on data on global market share. These systems encompass a variety of file system architectures, including NTFS and EXT4 providing a comprehensive basis for cross-platform forensic evaluation. To maintain experimental validity, all tests were conducted using standardized virtual machine configurations with minimal impact. Each operating system was tested under two RAM configurations, 4GB and 8GB, to examine the influence of system resources on the quality of forensic acquisition and malware behavior. VMware Workstation Pro 17 and Oracle VirtualBox was used to configure all virtual machines with identical virtualization settings, minimizing the introduction of platform-specific artifacts. The host machine ran on an Intel i5-7200U processor with virtualization support enabled, ensuring consistent performance and resource allocation under all experimental conditions.

Before introducing malware, each operating system was cleaned and installed using the default settings. Simulated user activities were performed, such as browsing the web, creating documents, opening media files, and making basic system changes.

Test data were added to each system, including text files of various sizes, and images to test how well forensic tools can detect them. MP3 audio files and MP4 videos were also included to create a realistic mix of digital artifacts.

The initial control data was collected by capturing memory dumps and disk images from the clean systems. FTK Imager was used for memory dumps on Windows and LiME was used for Linux.

Next, malware was introduced using cross-platform samples like remote access trojans (RAT) and ransomware. RATs such as CrossRAT.java, veryfun.exe, and icondance.exe were chosen for their compatibility with Windows and were downloaded

from public GitHub repositories like ¹ .

Ransomware samples, cerber.exe, cryptolocker.exe , and WannaCry, were also sourced from GitHub. These malware were run on isolated virtual machines, following a set infection protocol that included combinations of clean state ,single, double, and triple malware for Windows OS.

For Linux system, trojan malware introduced are Linux. Trojan.Dakkatoni, Gosh and brootkit which are sourced from malware bazaar.For ransomware, malware introduced are ransomexx, monti and gonnacry wahich are also sourced from Malware bazaar ². These malware were also run in linux OS with same set infection protocol as Windows OS.

3.1.1.1 Trojan Case Studies In Windows (8 total - one for each operating system Four 4GB case studies and fo ur 8GB case studies):

- Stage 0: (c0) clean state
- Stage 1: (c1) Single Trojan infection - RATs — CrossRAT.java
- Stage 2: (c2) Dual trojan infection - RATs — CrossRAT.java, veryfun.exe
- Stage 3: (c3)Triple trojan infection - RATs — CrossRAT.java, veryfun.exe, and icon dance.exe

3.1.1.2 Ransomware Case Studies in Windows (8 total - one for each operating system Four 4GB case studies and four 8GB case studies):

- Stage 0: (c0) clean state

¹URL's: <https://github.com/Da2dalus/The-MALWARE-Repo>

²URL :<https://bazaar.abuse.ch/>

- Stage 1: (c1) Single ransomware infection - cerber.exe
- Stage 2: (c2) Dual ransomware infection - cerber.exe, cryptolocker.exe
- Stage 3: (c3) Triple ransomware infection - cerber.exe, cryptolocker.exe , and WannaCry

3.1.1.3 Trojan Case Studies In Linux (8 total - one for each operating system Four 4GB case studies and four 8GB case studies):

- Stage 0: (c0) clean state
- Stage 1: (c1) Single Trojan infection - Linux.Trojan.Dakkatoni
- Stage 2: (c2) Dual trojan infection - Linux.Trojan.Dakkatoni, Gosh
- Stage 3: (c3) Triple trojan infection - Linux.Trojan.Dakkatoni, Gosh and brootkit

3.1.1.4 Ransomware Case Studies in Linux (8 total - one for each operating system Four 4GB case studies and four 8GB case studies):

- Stage 0: (c0) clean state
- Stage 1: (c1) Single ransomware infection - ransomexx
- Stage 2: (c2) Dual ransomware infection -ransomexx, monti
- Stage 3: (c3) Triple ransomware infection - ransomexx, monti and gonnacry

On the whole, our data collection involves 32 Windows dumps and 32 Linux dumps for our analysis. The introduction of malware followed a progressive approach.

3.1.2 Module 2: Reliable Evidence Acquisition

In stage 0, each malware type was run in clean state with new installation on virtual machine, the system was allowed to run for 10 min or beyond depending on the complexity, then forensic acquisition was performed. In stage 1, combinations of 1 malware instance from the same category were used, the system was run, and the acquisition was performed. In stage 2, 2 malware instances of the same category were used and the acquisition was performed. In Stage 3, All 3 malware was executed in isolated environments, and the final acquisition was performed to prevent unintended spread while ensuring realistic infection scenarios. During these stages, systems remained operational with regular user-like activities including web browsing, document file access, and application launches and closures. This approach allowed for the assessment of the persistence of evidence and the evolution of malware behavior over time in different operating systems. The following table 3.2 outlines the acquisition methods used for each operating system:

Operating System	Memory Dump Acquisition	Disk Image Acquisition
Windows 11	FTK Imager (.mem)	FTK Imager (E01 format)
Windows 10	FTK Imager(.mem)	FTK Imager (E01 format)
Ubuntu 24	Linux Memory Extractor (LiME) (.lime)	dd command (.dd format)
Ubuntu 20	Linux Memory Extractor (LiME)(.lime)	dd command (.dd format)

Table 3.2: Forensic Acquisition Methods by Operating System

For disk forensic acquisition, FTK Imager for Windows and dd for Linux was utilized. FTK Imager is primarily designed for Windows systems and does not natively work on Linux for both memory and disk acquisition. DD Method for Disk Imaging DD (Data Duplicator) is a command-line utility that copies data bit by bit from one location to another, making it ideal for forensic disk imaging. Creating exact duplicates of storage devices and preserving all data, including deleted files, unused space, and unallocated space, was our main data acquisition methodology.

For memory acquisition, platform-appropriate tools: FTK Imager for Windows and Linux Memory Extractor (LiME) for Linux were utilized. All memory dumps were acquired with both 4GB and 8GB RAM configurations to verify how memory size affects evidence recovery.

3.1.3 Data Analysis and Tool Selection

To test file recovery capabilities, 20 samples of each file type were distributed across the system. We added documents like Text documents (TXT, PDF); Images (JPG); and Audio files (MP3) and Video files (MP4) for ensuring all evidence formats are included in case studies. We ensured that half of these files were deleted and then permanently removed using platform-specific methods (Recycle Bin in Windows, rm command in Linux, while the other half remained in their respective folders. Recovery attempts were conducted using Autopsy 4.21.0 across all disk images, and recovered files were checked to see how many of them could be recovered and what complications and errors were faced in recovering data types. For disk forensic analysis, Autopsy 4.21.0 was used as the primary analysis platform, with several ingest modules enabled during analysis. We acquired all retrievable evidences that even includes Recent Activity, Hash Lookup for recovering all non-volatile data. To include volatile data, we conducted memory analysis with Volatility 3 framework was used for each operating system, utilizing platform-specific plugins Windows: malfind, pslist; Linux: linux_pslist, linux_netstat, linux_psscan, linux_filescan. The volatile evidence analysis recovered process identification, network connections, injected code identification, memory-resident artifacts, and hidden process detection.

3.2 Module 3: Reliable Analysis for Investigation

This experimental study provides a systematic set of best practices for evaluating forensic capabilities in diverse computing environments. By implementing various test scenarios across different operating systems with varying hardware configurations, the research enables empirical assessment of the effectiveness of forensic tools in real-world scenarios. We created analysis scenarios for Trojans and ransomware and how effectively it can be conducted across different time-based and resource-based environments. We conducted experiments with multiple time and resource-dependent analysis across different tools and ensured below-mentioned scenarios are achieved.

1. Malware-specific analysis challenge:

- Trojans may hide processes, inject code into legitimate processes, and establish persistent network connections that could be identified in memory forensics but may be difficult to detect in disk forensics
- Ransomware will likely create readily identifiable artifacts in both disk and memory forensics due to extensive file system modifications and encryption operations

2. Operating system-specific challenges may include:

- Windows systems exhibited complications with registry analysis and memory address translation
- Linux systems presented challenges with different kernel versions affecting memory structure mapping

In general, due to the extensive scope associated with Windows, our analysis focused primarily on the impact of incorrect cyber forensic environments on critical cyber investigations. The reliable analysis results are shown in Chapter 4.

Chapter 4

Evaluation and Effectiveness Assessment

4.1 Windows Case Study Findings: Impact of Trojans and Ransomware on Forensic Analysis

4.1.1 Trojan Infection Analysis

Analysis of memory dumps from Windows 11 and Windows 10 systems infected with trojans revealed notable differences in forensic visibility. Volatility 3 plugins `windows.pslist`, `windows.psscan`, and `windows.pstree` were utilized for process analysis. As shown in Table 4.1, `windows.pslist` identifies active processes by traversing the `_EPROCESS` list, while `windows.psscan` detects additional hidden or terminated processes by scanning memory pools. Hidden processes were identified when entries appeared in the `psscan` output but were absent from the `pslist` results. Suspicious memory regions were detected using the `windows.malfind` plugin.

Table 4.1 presents the number of processes identified by each plugin during the analysis of Windows 11 (8 GB) and Windows 10 (8 GB and 4 GB) systems infected with trojan malware.

Volatility plugin	C0 Clean state		C1 Trojan infection		C2 Trojan infection		C3 Trojan infection	
	8GB	4GB	8GB	4GB	8GB	4GB	8GB	4GB
Windows.pslist	161	x	142	x	139	x	151	x
Windows.psscan	143	x	141	x	139	x	152	x
Windows.pstree	162	x	33	x	139	x	171	x
Windows.malfind	23	x	23	x	20	x	27	x

Table 4.1: Windows 11 Trojan Malware - Volatility Plugin Output

We observed that the case study for Windows 11 with 4 GB RAM is missing, and we will address this issue in the “Complications and Errors” section related to case study execution. We have also noted that 4GB systems are sufficient for Windows 11¹ however, during our evaluation we did not receive reliable analysis and that is a major issue why we decide to mark the symbol ‘x’ in tables with Windows 11 analysis results. We primarily note that we used FTK Imager for memory dump acquisition and Volatility 3 for memory forensic analysis on Windows operating systems.

We began our analysis with the clean state (C0) by setting up a new computer system within a virtual environment and populating it with document-type files as described previously. After preparing the environment, we acquired the memory dump using FTK Imager and collected a disk image for subsequent disk forensic analysis.

For the C1 trojan infection state (CrossRAT), which involves Java-based malware, we installed Java in the virtual environment to enable the malware’s execution. During Volatility 3 analysis, we observed a process with the .java extension and the file name `mediamgrs.jar`. An alert notification appeared during execution and persisted until the C3 infection stage.

¹<https://support.microsoft.com/en-us/windows/windows-11-system-requirements-86c11283-ea52-4782-9efd-7674389a7ba3>

In the C2 trojan infection state, the virtual environment contained both Cross-RAT and `veryfun.exe` malware. At this stage, memory forensic plugins revealed filenames such as `memcompression`, `MoNotification`, and `javaw.exe`. We observed that `veryfun.exe` generated persistent alert errors and exhibited behavior indicative of OS corruption. Based on intervention during the case study, we determined that it is advisable to shutdown the system promptly, as malware continued to execute even after attempts to terminate the process using Task Manager, subsequently displaying the alert message: “FUN FUN FUN!!!”

In the final stage of C3 trojan infection, we introduced the `icondance.exe` malware into the environment already infected with two other malwares. This malware reorganized application icons on the desktop in a random pattern. As shown in Table 4.2, the number of processes increased by approximately 7–10% in the C3 stage compared to the clean state.

We observed a similar pattern in Windows 10 systems, where trojan infections in the virtual environment led to a 5–10% increase in process activity from the C1 to the C3 stages during our analysis of trojan malware behavior in Windows environments.

Volatility plugin	C 0 Clean state		C1 trojan infection		C2 trojan infection		C3 trojan infection	
	8gb	4gb	8gb	4gb	8gb	4gb	8gb	4gb
Windows.pslist	126	135	142	143	143	96	148	129
Windows.psscan	130	140	144	148	143	125	147	135
Windows.pstree	128	136	28	143	142	95	157	124
Windows.malfind	23	12	25	14	25	8	25	19

Table 4.2: Windows 10 trojan malware

Analysis: Based on these findings, we recommend prioritizing 8 GB Windows 10 and Windows 11 environments for memory forensics in trojan-infected systems to maximize evidence retention and analysis efficiency during cybercrime investigations, while acknowledging the potential but limited utility of 4GB systems in environments where hardware constraints require

their use.

4.1.2 Ransomware Infection Analysis

Ransomware infections created substantially different forensic artifacts compared to trojans. Using the core Volatility 3 plugins (`windows.pslist`, `windows.psscan`, `windows.pstree`), and `windows.malfind` for malware detection as shown in Table ??.

Volatility plugin	C0 Clean state		C1 ransomware		C2 ransomware		C3 ransomware	
	8gb	4gb	8gb	4gb	8gb	4gb	8gb	4gb
Windows.pslist	142	x	176	x	145	x	135	x
Windows.psscan	143	x	182	x	152	x	136	x
Windows.pstree	139	x	35	x	144		135	x
Windows.malfind	23	x	23	x	35	x	24	x

Table 4.3: Windows 11 ransomware malware

Volatility plugin	C0 Clean state		C1 ransomware infection		C2 ransomware infection		C3 ransomware infection	
	8gb	4gb	8gb	4gb	8gb	4gb	8gb	4gb
Windows.pslist	131	133	132	141	125	136	120	122
Windows.psscan	132	139	131	143	135	141	123	144
Windows.pstree	130	133	132	140	133	138	120	135
Windows.malfind	22	24	14	11	19	11	5	7

Table 4.4: Windows 10 ransomware malware

In our analysis of ransomware malware, we observed distinct patterns and behaviors compared to trojan infections. For the ransomware case study, we employed the same tools as in the trojan analysis, using FTK Imager for memory acquisition and Volatility 3 for memory forensics, and followed a similar procedure to acquire memory dumps across the C0 to C3 infection states.

To avoid contamination from previous malware infections, we created a new virtual environment for the clean state (C0). In this state, memory forensic analysis revealed no suspicious activity. In the C1 infection state, we introduced the `cerber.exe` ransomware. Volatility analysis identified suspicious activity, including error output with

machine-readable code patterns in memory offsets and anomalous entries within Task Manager.

During the C2 infection state, we deploy `cryptolocker.exe` ransomware. We observed that the malware began to replicate even after attempts to terminate the process via Task Manager, branching into processes with different PIDs and continuing its activity.

The most impactful malware in this study was `wannacry` in the state of C3 infection. Unlike the three previous malware samples that we examined, `wannacry` immediately encrypted files within the Windows system, altering document-type data files that had remained unchanged under previous malware infections. Our volatility-based memory forensic analysis revealed that, unlike trojans which typically result in an immediate increase in the number of active processes, ransomware infections led to a reduction in typical OS activities. Due to the encryption mechanisms of ransomware, we observed a 4–12% decrease in overall system activity compared to trojan infections during comparable infection states.

Analysis: Based on our findings, we recommend using 8GB systems over 4GB systems for memory forensics to achieve greater precision in investigations. The additional memory capacity not only improves artifact retention, but also accommodates system-level constraints such as page swapping, making it more effective for forensic analysis.

4.1.3 Disk Forensic Findings Across Windows Systems

Case studies involving trojan and ransomware infections revealed significant variations across Windows versions. We conducted the analysis using Autopsy 4.21.0, focusing on suspected encryption activities and extension mismatches, as well as identifying suspicious items, bad items, potential zip bombs, and artifacts within the Recycle Bin.

Analysis results	C0 Clean	C1 Tro- jan	C2 Tro- jan	C3 Tro- jan	C1 Ra- nsom- ware	C2Ra- nsom- ware	C3Ra- nsom- ware
Encryption Suspected	6	4	4	7	5	1	1
Extension Mismatch Detected	91	206	214	223	94	96	126
Suspicious items	97	199	219	231	99	97	127
Bad items	0	1	1	1	0	0	0
Interested Item (zip bomb)	0	1	1	1	0	0	0
Recycle bin	19	19	19	19	20	20	20

Table 4.5: Analysis Results of Various Infections on Windows 11 (8GB)

For the disk forensics analysis, we utilized Autopsy as our primary tool. As shown in Tables 4.12, 4.13, and 4.7, which present results for Windows 11 (8 GB) and Windows 10 (8 GB and 4 GB), we observed a slight increase in detected artifacts from the C1 to C3 malware infection states, with the exception of the C3 ransomware infection stage. When comparing Windows 11 to Windows 10, we found that Autopsy on Windows 11 performed better in detecting extension mismatches and suspected encryption activities.

Analysis:Based on these results, we conclude that Windows 11 demonstrates a stronger capability to resist and detect malware compared to Windows 10 within the context of our disk forensics analysis.

Analysis results	C0 clean state	C1 Tro- jan in- fec- tion	C2 Tro- jan in- fec- tion	C3 Tro- jan in- fec- tion	C1 Ra- nsom- ware	C2Ra- nsom- ware	C3Ra- nsom- ware
Encryption Suspected	4	3	2	2	11	9	8
Extension Mismatch Detected	254	261	260	265	255	383	215
Suspicious items	258	264	263	268	267	393	223
Bad items	0	0	0	0	0	0	0
Interested Item (zip bomb)	0	0	1	1	1	1	0
Recycle bin	19	19	19	19	19	19	19

Table 4.6: Analysis Results of Various Infections of windows 10 (8gb)

Analysis results	C0 clean state	C1 Tro- jan in- fec- tion	C2 Tro- jan in- fec- tion	C3 Tro- jan in- fec- tion	C1 Ra- nsom- ware	C2Ra- nsom- ware	C3Ra- nsom- ware
Encryption Suspected	6	9	7	9	9	12	12
Extension Mismatch Detected	258	283	267	268	253	253	256
Suspicious items	262	288	271	274	257	266	269
Bad items	0	0	0	0	0	0	0
Interested Item (zip bomb)	0	1	0	1	1	1	0
Recycle bin	19	19	19	19	19	19	12

Table 4.7: Analysis Results of Various Infections of windows 10 (4gb)

Findings from Case Studies in Windows and Trojan

Our observations indicate that trojan malware infections result in fewer instances of unwanted file encryption within the operating system compared to ransomware infections. Additionally, the number of running processes increases as the infection progresses in both malware categories. Specifically, in the case of ransomware, the number of running processes at the C1 infection stage was lower than at the C2 and C3 stages, reflecting the increasing encryption activity and system manipulation as the infection advances.

These findings suggest that system activity levels generally increase with the proliferation of malware within the operating system, whether from trojans or ransomware. An additional notable observation during our analysis was the occurrence of a ZIP bomb. A ZIP bomb (or decompression bomb) is a maliciously crafted compressed file (e.g., `.zip`, `.tar.gz`, `.7z`) designed to expand to an extremely large size up to terabytes of memory upon decompression, consuming excessive CPU, RAM, and disk space while originating from a file as small as 64 KB.

Furthermore, both trojan and ransomware infections exhibited tendencies to corrupt the boot system while interfering with the system’s self repair mechanisms during

the progression of the infection.

4.2 Linux Case Study Findings: Impact of Trojans and Ransomware on Forensic Analysis

4.2.1 Trojan Infection Analysis

4.2.1.1 Memory Forensic Findings Across Linux Systems

Examining memory dumps from various Linux environments infected with trojan malware revealed notable differences in forensic visibility. We conducted our analysis using the Volatility 3 framework with Linux specific plugins, including `linux.pslist`, `linux.pstree`, and `linux.pidhashtable`, to identify distinct malware behaviors. Additionally, we utilized the `linux.elf` plugin to detect suspicious or malicious ELF files during memory forensics.

Table 4.8 presents the number of processes identified by each plugin during the analysis of Ubuntu 24.04 and Ubuntu 20.04 systems with 8 GB and 4 GB RAM configurations infected with trojan malware.

Volatility plugin	C 0 Clean state		C1 trojan infection		C2 trojan infection		C3 trojan infection	
	8gb	4gb	8gb	4gb	8gb	4gb	8gb	4gb
<code>linux.pslist</code>	148	153	151	137	136	129	145	137
<code>linux.pidhashtable</code>	151	155	140	143	143	144	152	136
<code>linux.pstree</code>	147	153	149	91	136	121	150	139

Table 4.8: Ubuntu 24 trojan malware

For memory dump collection, we used LiME (Linux Memory Extractor), a kernel module designed to acquire full physical memory dumps from running Linux systems in a format compatible with Volatility 3. Since Volatility 3 does not provide built-in profiles for Linux operating systems as it does for Windows, we generated the necessary

Linux profiles prior to conducting our analysis. We began our case study with trojan malware infections as follows:

Table 4.9 clearly depicts that in clean state C0, memory forensics did not reveal suspicious file activity within the Linux system. During the state of C1 trojan infection, we deployed a malware sample referred to as `linux.trojan.dakkatoni`. Our analysis detected unusual files, including `/usr/lib/.../diskmanagerd` and `/tmp/.../diskmanagerd`, along with the execution of scripts such as `gcc4lef.sh`, visible on the system monitor. In the C2 trojan infection state, we identified multiple instances of suspicious file activity, such as:

```
(4321    0x7fab000    0x7faf000    /tmp/.dakkatoni)
```

These instances appeared under different PIDs, along with additional suspicious files with extensions such as `/tmp/helpdd`, `.txt`, and `.sh`. In the final state of C3 trojan infection, we attempted to deploy the `brookit` malware. However, the ELF file could not run due to an “Permission Denied” alert, and the associated ZIP archive was automatically deleted by the system a few minutes after loading. As a result, we did not observe any new suspicious activity during this stage of the investigation.

Volatility plugin	C 0 Clean state		C1 trojan infection		C2 trojan infection		C3 trojan infection	
	8gb	4gb	8gb	4gb	8gb	4gb	8gb	4gb
<code>linux.pslist</code>	155	135	147	143	127	151	141	132
<code>linux.pidhashtable</code>	149	143	149	152	138	151	153	140
<code>linux.pstree</code>	153	139	161	143	130	156	144	134

Table 4.9: Ubuntu 20 trojan malware

Analysis: Our analysis revealed that trojan malware infections exhibited similar behavioral patterns in both Linux operating systems examined. However, we observed a notable difference in the time required to collect memory dumps between the systems. Specifically, Ubuntu 24.04 required, on average, 5 to 8% longer to complete memory dump acquisition compared

to Ubuntu 20.04.

4.2.2 Ransomware Infection Analysis

Ransomware infections created substantially different forensic patterns compared to trojans. We have used the same procedure to take the memory dumps as in the Trojan infection case and the same volatility plugins for memory forensic analysis.

Volatility plugin	c 0 Clean state		C1 ransomware infection		C2 ransomware infection		C3 transomware infection	
	8gb	4gb	8gb	4gb	8gb	4gb	8gb	4gb
linux.pslist	144	134	154	112	x	x	x	x
linux.pidhashtable	150	138	151	134	x	x	x	x
linux.pstree	148	138	154	115	x	x	x	x

Table 4.10: Ubuntu 24 ransomware malware

We utilized **ransomexx** (C1), **monti** (C2), and **gonnacry** (C3) as the ransomware samples for our analysis. In the C0 clean state, memory forensic analysis using Volatility 3 revealed no suspicious activity. During the C1 infection state with **ransomexx**, we observed behavior similar to **wannacry** infections on Windows systems, where the ransomware generated an unidentified text file named **WHY-FILES-ARE-ENCRYPTED.txt** within the Videos directory on both Linux systems examined.

Following infection with the second ransomware, **monti** (C2), on top of **ransomexx**, the Ubuntu system began to exhibit corruption patterns consistent with **wannacry**, actively encrypting files, including documentation files such as **.txt** and **.pdf**, converting them into encrypted files **.monti** . During attempts to analyze memory dumps postinfection, Volatility 3 was unable to process the dumps, displaying Symbol/Kernel Profile Errors, specifically indicating an inability to determine kernel ASLR (Address Space Layout Randomization). A similar encryption process was observed during the C3 infection state with **gonnacry**, continuing the active encryption of files within the system as shown in tables 4.10 and 4.11.

Volatility plugin	c 0 Clean state		C1 ransomware infection		C2 ransomware infection		C3 transomware infection	
	8gb	4gb	8gb	4gb	8gb	4gb	8gb	4gb
linux.pslist	151	143	147	133	x	x	x	x
linux.pidhashtable	149	146	149	138	x	x	x	x
linux.pstree	151	153	161	140	x	x	x	x

Table 4.11: Ubuntu 20 ransomware malware

Our observations indicate that ransomware variants such as `monti` and `gonnacry` exhibit a significantly higher potential to harm systems compared to trojan malware. Ransomware infections result in a greater degree of system corruption, actively modifying system files in ways that can interfere with the memory acquisition process, whereas trojan infections generally do not cause such extensive file system alterations.

4.2.2.1 Disk Forensic Findings Across Linux Systems

For the disk forensic analysis of the Linux operating system, we have used Autopsy tool for analysis.

Analysis results	C0 clean state	C1 Tro- jan in- fec- tion	C2 Tro- jan in- fec- tion	C3 Tro- jan in- fec- tion	C1 Ra- nsom- ware	C2Ra- nsom- ware	C3Ra- nsom- ware
Encryption Suspected	5	7	9	9	6	1	2
Extension Mismatch Detected	238	245	254	251	271	264	303
Suspicious items	240	255	263	256	278	269	308
Bad items	0	0	0	0	0	0	0
Interested Item (zip bomb)	0	1	1	0	1	1	1
Recycle bin	19	19	19	19	19	19	19

Table 4.12: Analysis Results of Various Infections of Ubuntu 24 (8gb)

Based on our analysis across both system configurations (8GB and 4GB RAM) running the same Linux operating system, we found that variations in RAM capacity did not produce a substantial difference in the results of disk forensics.

Analysis results	C0 clean state	C1 Tro- jan in- fec- tion	C2 Tro- jan in- fec- tion	C3 Tro- jan in- fec- tion	C1- ranso- mware	C2- ranso- mware	C3- ranso- mware
Encryption Suspected	6	7	7	10	7	1	1
Extension Mismatch Detected	241	249	253	259	271	277	286
Suspicious items	244	254	261	266	274	280	293
Bad items	0	0	0	0	0	0	0
Interested Item (zip bomb)	0	1	1	0	0	1	1
Recycle bin	19	19	19	19	19	19	19

Table 4.13: Analysis Results of Various Infections of Ubuntu 24 (4gb)

As we can see, an increase of suspicious items and Extension Mismatch Detected item in analysis result can be found as the number of malware infections increases in both trojan and Ransomware malware.

Analysis results	C0 clean state	C1 Tro- jan in- fec- tion	C2 Tro- jan in- fec- tion	C3 Tro- jan in- fec- tion	C1- ranso- mware	C2- ranso- mware	C3- ranso- mware
Encryption Suspected	3	5	5	6	4	2	1
Extension Mismatch Detected	201	241	253	259	265	281	295
Suspicious items	209	248	256	264	271	281	298
Bad items	0	0	0	0	0	0	0
Interested Item (zip bomb)	0	1	1	1	1	1	1
Recycle bin	19	19	19	19	19	19	19

Table 4.14: Analysis Results of Various Infections of Ubuntu 20 (8gb)

Compared to trojan malware, ransomware infections in Linux systems exhibited an approximate 8 to 15% increase in suspicious activity. Trojan infections, by contrast, showed a 3 to 5% increase in activity as the number of trojan samples introduced into the system increased.

Analysis of the Recycle Bin revealed that, even after ransomware encryption, disk

Analysis results	C0 clean state	C1 Tro- jan in- fec- tion	C2 Tro- jan in- fec- tion	C3 Tro- jan in- fec- tion	C1- ranso- mware	C2 - ranso- mware	C3- ranso- mware
Encryption Suspected	3	2	4	6	4	2	1
Extension Mismatch Detected	194	235	241	243	265	282	289
Suspicious items	199	242	244	249	271	288	296
Bad items	0	0	0	0	0	0	0
Interested Item (zip bomb)	0	0	1	0	0	1	1
Recycle bin	19	19	19	19	19	19	19

Table 4.15: Analysis Results of Various Infections of Ubuntu 20 (4gb)

forensics remained effective in recovering deleted data artifacts from the Recycle Bin. However, attempts to recover files that had not been deleted prior to encryption were unsuccessful, particularly for `.txt` and `.pdf` formats, whereas video and audio files were more likely to be partially recoverable before malware induced corruption.

Additionally, both Windows and Linux environments demonstrated susceptibility to ZIP bomb data corruption during both trojan and ransomware attacks. The results of these findings are summarized in Tables 4.12, 4.13, 4.14, and 4.15.

4.3 Impact of System Memory on Forensic Artifacts

This research examined the impact of system memory configurations (4 GB versus 8 GB) on memory forensics and disk forensics across all operating systems tested. Table 4.16 presents the percentage differences in the artifacts of suspicious items identified when comparing the 4GB and 8GB configurations during triple malware infections (infection state C3) in stage 3. We used suspicious items as the basis for comparison, as autopsy flags these artifacts during disk forensics when files, data structures, or

artifacts appear potentially abnormal, malicious, hidden, or inconsistent with typical system activity. From Table 4.16, we observe that during trojan malware infections in stage C3, the number of suspicious items generally decreases when the system memory is reduced from 8GB to 4GB, with the exception of the Windows OS, where a slight increase is noted; however, this increase remains minimal. In the case of ransomware infections, both Windows and Linux systems exhibit a clear downward trend in the number of suspicious items detected as memory decreases. It is important to note that these results may vary, as different types of malware can affect operating systems in different ways. We will address these variations in the “Complications and Errors” section. The change in percentage of data recovered is less than 15% however, even 15% of evidence might be useful for forensic evidence, hence using 8GB RAM is recommended at minimum for forensic investigation.

Table 4.16: Difference in Forensic Visibility: 4GB vs. 8GB Systems

Operating System	Disk forensic - Trojan		Disk forensic - Ransomware	
	8GB	4GB	8GB	4GB
Windows 11	231	-	127	-
Windows 10	268	274 (+2%)	233	269 (+13%)
Ubuntu 24.04	263	261(-0.5%)	308	298(-3%)
Ubuntu 20.04	264	249(-5%)	298	296(-1%)

4.4 Persistence Differences: Trojans vs. Ransomware

Through extensive case studies conducted on both Windows and Linux systems, progressively increasing the number of malware samples from the C0 to C3 stages, we observed that in trojan infections, unique signatures of the three trojans remained identifiable at the C3 stage during memory forensics on both operating systems. These trojans continued to run and remained deeply embedded within the system files, exhibiting higher persistence compared to ransomware. In contrast, our memory forensic

analysis of ransomware infections indicated that ransomware samples act independently rather than in conjunction, due to their encryption behavior. Ransomware often reencrypts files that have already been encrypted, a behavior observed notably with **gonnacry** and **wannacry** samples. Disk forensics, however, consistently revealed instances of all malware samples, whether trojan or ransomware, identifiable within the file types section of Autopsy. Based on these observations, we conclude that disk forensics offers a comparative advantage over memory forensics in terms of comprehensive malware detection across infection stages.

4.5 Time Taken for Forensic Acquisition Data

Table 4.17: Time taken for memory dump acquisition for Trojan Malware

Operating System	C1	C2	C3
Windows 11	5.14	6.26	7.28
Windows 10	3.14	4.08	5.54
Ubuntu 24.04	2.21	2.54	3.11
Ubuntu 20.04	1.46	2.38	3.41

Table 4.18: Time taken for Disk Image acquisition for Trojan Malware

Operating System	C1	C2	C3
Windows 11	26.08	29.41	34.28
Windows 10	29.11	55.52	38.26
Ubuntu 24.04	22.47	28.34	15.04
Ubuntu 20.04	17.56	34.17	28.39

The time required by forensic tools, such as FTK Imager for Windows OS and LiME (Linux Memory Extractor) and dd (disk duplicator) for Linux OS, as the number of malware samples increases, is a critical factor in understanding how malware infections impact data acquisition in both memory and disk forensics.

Table 4.19: Change in Time taken for Forensic data acquisition for Trojan malware in percentage

Operating System	C1→C2 (%)	C2→C3 (%)	C1→C2 (%)	C2→C3 (%)
	Memory forensic		Disk Forensic	
Windows 11	21.86	16.28	12.77	16.54
Windows 10	29.94	35.78	90.73	31.4 ↓
Ubuntu 24.04	14.93	22.44	26.10	46.94 ↓
Ubuntu 20.04	63.01	43.28	94.57	16.93 ↓

Tables 4.17, 4.19 illustrate the progressive increase in acquisition time across infection stages (C1, C2, C3) for triple infections on 8 GB systems, with time measured in minutes. Focusing first on memory dump acquisition, we observe that as the number of malware samples increases, the time required for forensic tools such as FTK Imager and LiME to complete data acquisition also increases. Table 4.18 presents the time required for disk image acquisition during trojan malware infections on 8 GB systems. Disk images were captured using external storage devices, such as hard drives or USB drives, to store the acquired images, with time recorded in minutes. From the data, we observe that the average time required to acquire a disk image ranges from approximately 25 to 35 minutes, with a gradual increase corresponding to the number of malware infections. Irregularities, such as in the C2 stage on Windows 10 and the C3 stage on Ubuntu 24.04, are attributable to hardware changes (e.g., different external drives) and system corruption caused by active malware during acquisition. Analysis of data indicates that in memory forensic acquisition, there is an approximate 30% increase in acquisition time from C1 to C2 and a 25% increase from C2 to C3. In disk forensics, we observe a variable increase in acquisition time across operating systems due to the nature of full-disk image acquisition. The following factors contribute to these variations:

- External storage devices, such as USB drives and hard disks, have varying upload speeds and compatibility issues between operating systems, leading to fluctua-

tions in acquisition time. A decrease in acquisition time from C2 to C3 may occur due to these factors and is indicated as ($\downarrow$) in the tables.

- Disk image acquisition generally takes significantly longer than memory dump acquisition. During this process, the operating system may experience crashes or freezes, resuming operation after a delay, thereby increasing the overall acquisition time.

4.6 Errors and Complications in Forensic Analysis

During our comprehensive analysis across operating systems, we encountered various errors and complications that impacted forensic processes and outcomes. These challenges arose across both Windows and Linux environments and were influenced by factors related to the host machines used for case studies as well as the virtual machine platforms employed, including VMware Workstation and Oracle VirtualBox.

4.6.1 Operating System complications

- Many types of malware fail to execute due to compatibility issues between the operating system and the malware's executable format. For example, Linux systems support ELF (Executable and Linkable Format) files, while Windows systems support `.exe` files. This disparity makes it challenging to generalize how the same malware type affects different operating systems.
- On Windows 11, after acquiring forensic data on an 8 GB RAM configuration for trojan and ransomware samples, the virtual machine displayed errors such as triggering the automatic repair process. Even after several hours, the Windows 11 VM host failed to restart. Attempts to resolve the issue using a new ISO image

and switching to Oracle VirtualBox were unsuccessful, with the Windows 11 VM failing to start in both VMware Workstation and VirtualBox. Consequently, we could not acquire forensic data for the 8 GB configuration on Windows 11, an issue that did not occur on Windows 10 VMs.

- The host system occasionally experienced sudden crashes during forensic acquisition and analysis, manifesting as a blank white screen or system lockup/freezes.
- Extended periods of continuous operation without restarting the host machine led to delays when opening virtual machines on both VMware Workstation and VirtualBox. Ubuntu 24.04, in particular, exhibited signs of system overload, failing to open immediately as Ubuntu 20.04 did. In these cases, restarting the entire host machine was necessary to resume work on Ubuntu 24.04, whereas for Ubuntu 20.04, closing and reopening the VM itself was sufficient.

4.6.2 Memory Forensic Complications

- Volatility is available in two versions: Volatility 2 and Volatility 3. Since Volatility 2 does not support Windows 11, we used Volatility 3 for all Windows 11 and 10 memory forensic analyzes.
- When analyzing memory dumps from Windows 11, Volatility may display symbol table and profile errors. If using an older version, such as Volatility 2.7.0, upgrading to the latest version is recommended to resolve these errors.
- It is essential to ensure that all required Python libraries are installed in the environment; otherwise, `ModuleNotFoundError` may occur. Additionally, specifying the correct path to the memory dump is critical, as incorrect paths will result in `FileNotFoundError`.

- Malware infections, particularly ransomware, can corrupt the system's OS symbol table. In our Linux OS case studies, we observed that at the C2 and C3 infection stages, forensic analysis often failed due to malware induced encryption and kernel corruption, resulting in corrupted memory dumps. Consequently, Volatility was unable to process these memory dumps for analysis.

4.6.3 Disk Forensic Complications

- In disk forensics, autopsy requires an average of 2 to 3 hours to process a given disk dump. During this process, it is recommended that the host machine remain idle, as concurrent activities may cause the application to freeze or enter a processing loop, necessitating a complete restart of the analysis to obtain accurate forensic data.
- For Windows systems, the use of FTK Imager during disk dump acquisition is recommended. This tool can export a structured list of files and directories from an acquired image to Excel compatible formats (CSV or `.xlsx`), facilitating forensic reporting, timeline reconstruction, and the identification of suspicious files, such as those with unusual extensions, large sizes, or hidden attributes.
- For Linux systems, selecting the correct file system path (e.g., `/dev/sda`) is essential during acquisition. Failure to specify the correct disk path will prevent Autopsy from accurately reading and processing the disk dump.
- Severe disk corruption or ransomware overwrites can hinder Autopsy's ability to successfully carve and recover files from disk images.
- Malware often hides in directories such as `/tmp/.xyz` or `/var/.abc`. These hidden files may be overlooked unless explicitly searched during analysis.

- Files with extensions such as `.jpg` or `.txt` may, in reality, be executable or encrypted files. Malware infections may also generate duplicate files containing random data while sharing identical hash values, complicating file integrity analysis.

Limitations and Future Research Directions

While this research provides comprehensive information on the behavior of malware in Windows and Linux operating systems, several limitations must be acknowledged. The use of virtual machines, while essential for consistency and isolation, may not fully replicate the behavior of physical hardware, particularly in areas such as memory management and storage performance. Future studies should consider testing on physical systems to validate these findings. Furthermore, while the selected malware samples represent common cross-platform threats, they do not capture the full diversity of malware types, particularly those designed for specific operating systems. Future research should incorporate a broader range of targeted malware samples to enhance generalizability. Reliance on Volatility 3 and Autopsy 4.21.0 introduces potential tool-specific biases. Future work would benefit from comparing results across multiple forensic tools to distinguish genuine platform behaviors from tool-induced artifacts, thus enhancing the robustness and reliability of forensic analysis outcomes.

Chapter 5

Conclusion

The research presents a systematic empirical analysis of reliability challenges in cyber forensic investigations across Windows and Linux environments. Using a controlled experimental best practices with virtual machines to ensure hardware consistency, the study evaluated four operating systems under varied malware infection scenarios, system configurations, and acquisition timelines. The findings demonstrate that factors such as operating system architecture, memory resources, malware type, and the timing of forensic acquisition significantly impact evidence visibility and degradation. By incorporating both memory and disk forensics using standardized procedures and platform specific tools, the study ensured consistent and credible cross-platform evaluations. Key contributions of this thesis include:

- Comprehensive cross-platform empirical forensic analysis under uniform malware conditions.
- Identified memory limitations affecting investigation quality, supporting the recommendation of a minimum of 8 GB RAM for reliable forensic analysis.
- Identification of distinct forensic patterns in trojan and ransomware infections

using empirical study.

- Development of empirically grounded timelines for evidence degradation during malware infections.
- Practical documentation of tool specific error behaviour across platforms to support forensic practitioners.

Additionally, the results suggest that encrypted content demonstrates resilience against ransomware attacks, supporting encryption as a defensive strategy for system protection during malware incidents. While this research provides comprehensive insights, several limitations should be noted. The use of virtual machines, although essential for experimental control, may not fully replicate the behavior of physical hardware, particularly in areas like memory management and storage performance. Additionally, while the selected malware samples represent common cross-platform threats, they do not capture the complete diversity of malware, especially those specifically targeting individual operating systems. The reliance on Volatility 3 and Autopsy 4.21.0 may introduce tool specific biases, warranting comparative studies using additional forensic tools to isolate platform behaviors from tool induced artifacts. Future work will extend this analysis to macOS environments to complete the cross-platform study and evaluate additional types of targeted malware to enhance generalizability. Furthermore, we plan to explore the role of AI in forensic analysis while emphasizing the limitations of AI in replacing human expertise in cyber forensics, ensuring that human judgment remains central in critical forensic investigations. Overall, this thesis underscores that effective and credible cyber forensic investigations require a combination of appropriate tool selection, timely evidence collection, and investigator expertise, all of which are crucial for maintaining reliability and accuracy in digital forensics. The findings and methodologies established through this research provide a foundational reference for

advancing systematic and reliable practices in the field of cyber forensic investigations.

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